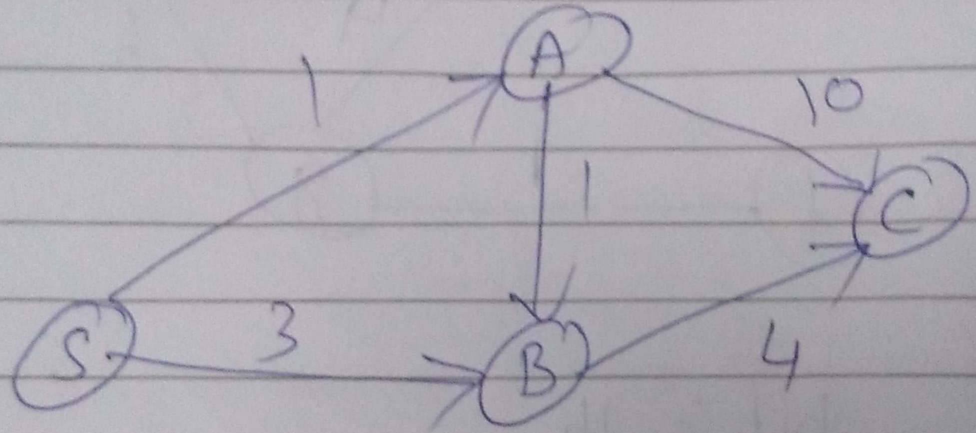
Dijkstra - Shortest path

Given a weighted graph G, and a source vertex, S, the algorithm returns the shortest path from S to any of the other vertices



Explored	Unexplored
	S^0 A^∞ B^∞ C^∞
S^0	A^{1S} B^{3S} C^∞
S^0 A^{1S}	B^{2A} C^{11A}
S^0 A^{1S} B^{2A}	C^{6B}
S^0 A^{1S} B^{2A} C^{6B}	

From S,

A : $S \to A = 1$

B : $S \to A \to B = 2$

C : $S \to A \to B \to C = 6$